A Theory of International Boycotts

Abdoulaye Ndiaye

New York University*

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Abstract

This paper provides the first systematic analysis of international boycotts within a neoclassical Ricardian framework. I establish a general equivalence result that equilibrium outcomes with boycotts can be analyzed through either the fixed factor demands or fixed expenditures of the boycotting coalition, thereby clarifying the economic mechanisms behind boycotts. Optimal boycotts (i) take the form of prohibitions on importing foreign goods unless foreign comparative advantage is sufficiently large, and (ii) boycotters optimally increase labor supply to spend more on domestic goods. International boycotts are equivalent to certain forms of government-imposed sanctions or voluntary export restrictions, although their effectiveness is diminished by leakage effects from nonboycotting consumers. Thus, while boycotts share key similarities with conventional trade policy instruments, they introduce complexities that position them as distinct tools of geoeconomic influence.

1 Introduction

In international trade and geopolitical relations, boycotts have emerged as a tool of market influence, akin in some respects to government-imposed sanctions and tariffs. While governments often employ sanctions and tariffs to exert political or economic pressure, boycotts reflect the collective choice of a coalition of households.

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International boycotts raise several questions: What are boycotts in a neoclassical economic sense? How should members of a boycotting coalition allocate their consumption and labor supply to achieve their objective? How do international boycotts relate to and differ from government-imposed sanctions or import tariffs?

Historically, boycotts have been used in social (Weems, 1995) and political (Kam and Deichert, 2020) movements. Economists, however, have not given extensive attention to the economic consequences of boycotts. The aim of this paper is to study the economics of international boycotts in a neoclassical Ricardian model. To my knowledge, this is the first treatment of consumer boycotts using the workhorse frameworks of mainstream economics.

In my baseline model à la Dornbusch et al. (1977), firms in two countries, Home and Foreign, produce and export a continuous set of differentiated goods ranked in decreasing order of relative productivity or comparative advantage for the Home country. An exogenous share of domestic consumers form a boycott coalition. These boycotters face a strategic dilemma: namely, they individually make consumption and labor supply decisions while selecting *among choices that hold fixed excess demand for each factor* while maximizing their welfare. I define our notion of competitive equilibrium (CE) with boycotters with fixed excess demand and show as my main result that this CE notion is equivalent to one where boycotters make utility-maximizing consumption and labor supply decisions *among choices that hold fixed their total expenditure for each country's goods.*

This equivalence result is general and applies beyond the DFS model to multi-country, multi-good settings with heterogeneous household economies featuring constant-returns-to-scale (CRS) production and a single labor factor per country. Intuitively, under these conditions, equilibrium factor prices depend solely on wages, allowing a direct correspondence between boycotters' excess factor demands and their expenditures on each country's goods.

Therefore, as a coalition, boycotters need to set the total budget they spend on each country to minimize the welfare of Foreign households while satisfying their individual participation constraints. This problem can be analyzed from the consumption and labor supply perspectives.

On the consumption side, boycotters allocate their demand to each good, given fixed expenditures on each country's goods. For any CE with boycotters having positive expenditure on domestic goods, the set of completely boycotted Foreign goods is nonempty. This result highlights that the usual notion of a boycott, a blanket prohibition of certain goods, is optimal to penalize Foreign welfare. In addition, every optimal boycott takes the form of a cutoff in the space of foreign comparative advantage; boycotters only import goods for which Foreign's comparative advantage is sufficiently large.

The labor supply side of boycotts is equally important when labor can be supplied inelastically. Optimally, a boycotter holds fixed factor demands by working more than nonboycotters and spending all of her earnings on domestic goods. Thus, a boycotter's labor supply decision should be undistorted from the perspective of domestic spending.

In relation to trade flows, expanding the set of boycotted foreign goods directly reduces imports by lowering the share of goods purchased abroad and indirectly boosts domestic wages through increased demand for domestic goods. Although higher domestic wages encourage imports, the direct effect dominates, resulting in a net reduction in imports. Concurrently, the expanded boycott reduces the equilibrium cutoff for exported domestic goods, thus decreasing exports to restore trade balance. This mechanism highlights an equivalence between a comprehensive consumer boycott and a voluntary export restriction (VER), reminiscent of the Lerner symmetry (Lerner (1936) and Costinot and Werning (2019)), which equates the effects of import tariffs and export taxes.

The presence of nonboycotting consumers reduces the effectiveness of a boycott through leakage effects. Specifically, the boycott indirectly raises domestic wages by shifting demand toward domestic goods, making foreign goods relatively cheaper for nonboycotters. Consequently, nonboycotters increase their imports from abroad, partially offsetting the boycott's intended impact. This leakage, consistent with existing literature on optimal unilateral consumer actions motivated by altruistic preferences (see Kaufmann et al. (2024) and Trammell (2025)), weakens the boycott's effectiveness. To maintain the same pressure on foreign welfare, boycotters thus need to target a broader set of goods as the share of nonboycotters increases.

In comparing full boycott coalitions with trade sanctions, a consumer boycott by all domestic households can be replicated by good-specific import tariffs. In particular, in a scenario where every consumer boycotts a range of foreign goods beyond the free trade equilibrium cutoff, one can construct tariffs sufficiently high to prevent imports of exactly the same goods. Both the boycott and tariff scenarios result in identical equilibrium allocations and trade flows because they produce the same shifts in relative prices and trade balance.

Comparing boycotts with linear import tariffs clarifies their equilibrium effects. Unlike boycotts, linear tariffs directly increase the cost of foreign goods, shifting the relative productivity curve and creating a set of nontraded goods. Boycotts, however, do not alter comparative advantage but instead operate through implicit income transfers from Foreign to Home due to geographic differences in consumer preferences. These transfers shift only the balanced trade curve, leading to higher domestic wages and reduced imports without creating nontraded goods in the DFS setting. Consequently, Foreign consumers lose access to efficiently produced Home goods, while domestic consumers benefit through improved terms of trade.

Extending the model highlights several important considerations regarding substitution elasticities, good/firm-specific boycotts, and strategic retaliation. Higher substitution elasticities improve the effectiveness of boycotts, as boycotters find it easier to switch to alternative products. When boycotts target specific firms due to political or social animus, they affect terms of trade depending on whether the targeted firms primarily produce imported or exported goods. Additionally, considering retaliation introduces strategic interaction between countries, creating incentives akin to tariff wars.

1.1 Literature Review

This paper contributes to the economic literature by analyzing international boycotts through the lens of a neoclassical Ricardian trade model, drawing connections to existing research on trade policies, economic sanctions, non-individualistic consumer behavior, and empirical studies on the trade effects of political conflict and international boycotts.

Trade Policies, Comparative Advantage, and Taste Shocks. The paper most closely related to mine is Opp (2010), which shows that the optimal import tariff in a DFS model is uniform, leading to a shift in imported varieties along the comparative advantage threshold and wage effects. My contribution is to derive analogous results on the optimal targeting of foreign varieties in a second-best scenario involving partial trade bans imposed by a subset of consumers.

Costinot et al. (2015) characterize optimal terms-of-trade manipulation in Ricardian settings, finding that optimal trade policies involve uniform tariffs on imports but heterogeneous taxes on exports. While their analysis encompasses more general preferences than the setting considered here, their problem is amenable to powerful public finance and Lagrangian techniques. In contrast, in my study of boycotts, focusing on consumption sets and labor supply choices, Lagrangian techniques find their limits at the CE equivalence result for the general formulation of my problem. To further characterize the optimal set of boycotted goods, I rely on methods involving measurability conditions and proofs by contradiction, techniques that could be valuable in other studies of optimal policies without direct price instruments.

Additionally, participating in a boycott can be viewed as experiencing a preference shock favoring home-produced over foreign-produced goods for a specific subset of consumers. Although preference shocks have been extensively studied in international trade and macroeconomic literatures (see, e.g., Stockman and Tesar (1990), Pavlova and Rigobon (2007), and Bai and Ríos-Rull (2015)), our equivalence between equilibria with fixed excess demands and equilibria with fixed expenditures, as well as the optimal structure of targeted boycotts as prohibitions on specific goods, represent novel results within the context of my analysis

Economic Sanctions. A burgeoning literature has emerged examining the economic impacts of sanctions and geopolitics in international trade (e.g., Grossman and Helpman (1994), Antràs and i Miquel (2011), Traiberman and Rotemberg (2023), Clayton et al. (2023); see also Antràs (2024) for a recent survey). In this literature, the closest paper to mine is Becko (2022), which draws a connection between trade sanctions and terms-of-trade manipulations. Similarly, Itskhoki and Ribakova (2024) analyze how various types of sanctions–including trade, financial, and payment restrictions–affect the Russian economy. Leveraging Arkolakis et al. (2012)'s welfare formulas, they find that the effect of trade sanctions can be measured by their impact on foreign imports. In contrast, in my paper, boycotters perform terms-of-trade manipulations, but the effectiveness of their actions can be weakened by nonboycotters' response to price adjustments.

Socially Responsible Consumers. This paper also contributes to the literature on market consequences of socially responsible behavior by price-taking consumers (e.g., Dewatripont and Tirole (2024), Aghion et al. (2023)). Related work by Kaufmann et al. (2024) and Trammell (2025) study optimal unilateral behavior by altruistic consumers who are not price-takers but rather internalize how their own actions will affect market outcomes in equilibrium. My modeling approach occupies an intermediate position between these two strands: individual boycotters remain price-takers but make their consumption and labor supply choices *among*–a constraint–those that hold fixed the total expenditures of the boycott coalition on each country's goods. This approach allows me to show leakage effects driven by nonboycotters' responses to equilibrium price changes while preserving the tractability of standard competitive equilibrium analysis.

International Boycotts and Political Conflict. This paper is motivated by empirical research on international boycotts and the broader literature documenting the effects of political conflicts on trade patterns (Iyoha et al. (2024); Gopinath et al. (2025) show empirically that trade flows often align along geopolitical boundaries). Specifically, the "surrogate" international boycotts I study are driven not by direct misconduct of firms but by geopolitical tensions between countries, where firms are targeted as representatives of their host countries. Prominent examples include the resurgent calls for boycotts of firms in and doing business with Israel during the Israel–Gaza war and, the US boycott of French products following France's opposition to the

Iraq war (Ashenfelter et al., 2007; Chavis and Leslie, 2009; Pandya and Venkatesan, 2016). Additional examples are the Arab League's boycotts of Danish products in 2005 and of French products in 2015 (Ammitzbøll and Vidino, 2007; Mirza et al., 2020). Moreover, government involvement can amplify international boycotts: Bohman and Pårup (2022) finds evidence that nearly one-third of all Chinese consumer boycotts have state support, suggesting that international boycotts that I study through a neoclassical framework serve as significant instruments of contemporary statecraft.

Outline. The rest of the article is organized as follows. Section 2 describes our baseline DFS model and presents all results in this setting. Section 3 generalizes our main result and discusses extensions of our analysis. Section 4 offers some concluding remarks. All formal proofs can be found in the Appendix.

2 A Benchmark DFS Model

I construct a theoretical framework based on the Dornbusch et al. (1977) model of international trade and study the role of consumer boycotts.

2.1 Environment

Two countries, namely, Home (H) and Foreign (F), engage in trade over a continuum of goods indexed by $j \in [0, 1]$.

Production: The model's production side is characterized by a single input, labor l, with a production function that exhibits a constant marginal product of labor for each good j. This marginal product is country specific, with $z_H(j)$ for Home and $z_F(j)$ for Foreign. Home's productivity in producing good j relative to Foreign's is given by $Z(j) = \frac{z_H(j)}{z_F(j)}$. Goods j are ordered such that the relative productivity of Home is decreasing. Total labor endowments are denoted by l_H and l_F for Home and Foreign, respectively, and labor moves freely across the production of different goods j within a country. Markets are perfectly competitive, and labor is the sole determinant of production costs, leading to equilibrium wages w_H and w_F in Home and Foreign, respectively.

I distinguish between regular consumers in Home or Foreign, who do not participate in the boycott, and domestic boycotters, who boycott goods from Foreign.

Regular Consumers: Regular consumers in both countries are atomistic and derive utility from a diverse set of goods irrespective of provenance, and their preferences are represented by a Cobb–Douglas utility function, expressed as:

$$U(\mathbf{q}^i) = \int_0^1 s(j)\log(q^i(j))dj \tag{1}$$

where $q^i(j)$ denotes the quantity of good j consumed by a regular consumer in country i and $s(j)$ signifies the expenditure share on good j, which sums to 1 across all goods. I denote by $S(j) = \int_o^j s(j')dj'$ the fraction of income spent on goods with index value below j, with $S(1) = 1$. The budget constraint of regular consumers in country $i \in \{H, F\}$ is:

$$\int_0^1 p(j)q^i(j)dj = w_i l_i \tag{2}$$

where $p(j)$ is the equilibrium world price of good j.

Given wages and prices, regular consumers make demand choices to maximize their utility (1) subject to their budget constraint (2). Since each good is specific to a particular producer and there is a unique factor, labor, the non-substitution theorem – See (Samuelson, 1951; Stiglitz, 1970), and (Acemoglu and Azar, 2020; Toda and Walsh, 2024) for recent treatments – implies that prices are uniquely determined by the supply side of the economy $p(j; \{w_H, w_F\})$. Similarly, the demands of the agents are unique given prices. Therefore utilities of all nonboycotters are determined by the vector of world wages. Let $U^H(\{w_H, w_F\})$ and $U^F(\{w_H, w_F\})$ denote these utilities for Home and Foreign regular consumers, respectively.

Consumer Boycotters: An exogenous share λ of domestic consumers derive utility from the same Cobb–Douglas utility function (1). Given wages, and prices, they maximize their utility subject to their budget constraint but choose consumption among goods that hold fixed excess demand for domestic and Foreign labor at some values $(\overline{L}_H, \overline{L}_F) \in [0, l_H] \times [0, l_F]$. Formally, boycotting consumers' problem is:

$$U^B(\{w_H, w_F, \overline{L}_H, \overline{L}_F\}) = \max_{(\mathbf{q}_H^B, \mathbf{q}_F^B)\in X} \int_0^1 s(j)\log(q_H^B(j) + q_F^B(j))dj \tag{3}$$

$$\text{s.t:} \quad \int_0^1 p_H(j)q_H^B(j)dj + \int_0^1 p_F(j)q_F^B(j)dj = w_H l_H \tag{4}$$

$$X = \left\{(\mathbf{q}_H^B, \mathbf{q}_F^B) : \quad \left(l_H - \lambda\int_0^1 \frac{q_H^B(j)}{z_H(j)}\,dj, l_F - \lambda\int_0^1 \frac{q_F^B(j)}{z_F(j)}\,dj\right) = (\overline{L}_H, \overline{L}_F)\right\} \tag{5}$$

In the rest of this paper, I define a CE with boycotters as follows.

Definition 1 (Competitive Equilibrium with Boycotters that Hold Fixed Excess Demand)**.** *A CE with boycotters corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule* *of world prices* $\boldsymbol{p}$, *and domestic prices* $\{\boldsymbol{p}_H, \boldsymbol{p}_F\}$, *consumption schedules* $\{\boldsymbol{q}^H, \boldsymbol{q}^F, \boldsymbol{q}^B\}$, *excess demand for labor inputs* $\left(\overline{L}_H, \overline{L}_F\right)$, *and output schedules that satisfy:*

1. *Firm optimization: price equal firms' marginal cost,*
2. *Regular consumer utility maximization: domestic and regular consumers maximize* (1) *subject to* (2),
3. *Consumer boycotter optimization: domestic boycotters maximize* (3) *subject to* (4), *and* (5).
4. *Labor market clearing in each country*

$$\int_0^1 \frac{\left((1-\lambda)q^H(j) + q^F(j)\right)}{z_H(j)} dj = l_H - \lambda \int_0^1 \frac{q_H^B(j)}{z_H(j)} dj = \overline{L}_H, \tag{6}$$

$$\int_0^1 \frac{\left((1-\lambda)q^H(j) + q^F(j)\right)}{z_F(j)} dj = l_F - \lambda \int_0^1 \frac{q_F^B(j)}{z_F(j)} dj = \overline{L}_F. \tag{7}$$

The Boycotting Coalition's Problem: The boycotting coalition chooses the vector of excess demands $\left(\overline{L}_H, \overline{L}_F\right)$ that gives the highest utility for the representative boycotter among CEs that provide utility to Foreign consumers below a certain level v^F.[1]

Definition 2 (Boycotting Coalition's Problem)**.** *The boycotting coalition's problem is*

$$\max_{\left(\overline{L}_H, \overline{L}_F\right)} U^B(\{w_H, w_F, \overline{L}_H, \overline{L}_F\}) \tag{8}$$

$$\text{s.th: } U^F(\{w_H, w_F\}) \leq v^F \tag{9}$$

and conditions (1)-(7) *hold.*

2.2 An Equivalence Result

I introduce some notation to further characterize CEs with boycotters. Denote by $\mathbf{q}_H^B = \mathbf{q}_H^B + \mathbf{q}_F^B$ the decomposition of boycotting consumers' demand for Home and Foreign goods. Let $E(\mathbf{q}_i^B) \equiv \lambda \int_0^1 p_i(j) q_i^B(j) dj$, for $i \in \{H, F\}$, be the total expenditure of boycotters on Home and Foreign goods, respectively, where $p_i(j)$ is the domestic price of each good j in country i, which can be distinct from world prices.[2]

Boycotters could make consumption choices while keeping their overall expenditure on Home and Foreign goods fixed. The following definition formalizes the notion of CE under this alternative choice set for boycotters.

[1] This problem is equivalent by duality to the problem of the boycotting coalition of minimizing Foreign's utility subject to achieving a certain level of utility for themselves and the other equilibrium conditions.

[2] By boycotting a set of Foreign goods that would not have been produced at Home otherwise because of Home's lack of comparative advantage, consumer boycotters can create demand for these to be produced domestically, albeit at a less competitive price $p_H(j)$.

Definition 3 (Competitive Equilibrium with Boycotters with Fixed Expenditures). *A CE with boycotters with fixed expenditures corresponds to a pair of wages, $\{w_H, w_F\}$, a schedule of world prices $\boldsymbol{p}$, and domestic prices $\{\boldsymbol{p}_H, \boldsymbol{p}_F\}$, consumption schedules $\{\boldsymbol{q}^H, \boldsymbol{q}^F, \boldsymbol{q}^B\}$, expenditures $\left(\overline{E}_H, \overline{E}_F\right)$, and output schedules that satisfy from optimization, regular consumer utility maximization (1), (2), labor market clearing in each country (6), (7), and consumer boycotter optimization (3) (4), and (10) where the fixed excess demand choice set is replaced by the fixed expenditure for each country goods choice set*

$$X = \left\{\left(\boldsymbol{q}_H^B, \boldsymbol{q}_F^B\right): \quad \left(E(\boldsymbol{q}_H^B), E(\boldsymbol{q}_F^B)\right) = \left(\overline{E}_H, \overline{E}_F\right)\right\}. \tag{10}$$

The following proposition provides an equivalence that links the two notions of CE, whether the with boycotters keeping excess demand for labor inputs fixed or holding fixed their overall expenditure for each country's goods.

Proposition 1 (Equivalence of CE with Boycotters with Fixed Excess Demand to CE with Boycotters with Fixed Expenditure). *A tuple $\{w_h, w_F, \boldsymbol{p}, \boldsymbol{p}_H, \boldsymbol{p}_F, \boldsymbol{q}_H, \boldsymbol{q}_F, \boldsymbol{q}_B, \overline{L}_H, \overline{L}_F\}$ is a CE with boycotters with fixed excess demands if and only if $\{w_h, w_F, \boldsymbol{p}, \boldsymbol{p}_H, \boldsymbol{p}_F, \boldsymbol{q}_H, \boldsymbol{q}_F, \boldsymbol{q}_B, e\left(\overline{L}_H\right), e\left(\overline{L}_F\right)\}$ is a CE with boycotters with fixed expenditures, with $e\left(\overline{L}_i\right) = w_i\left(l_i - \overline{L}_i\right)$ for $i \in \{H, F\}$.*

This proposition is general, as we show in Section 3, and states that boycotters ultimately need to make a choice on the overall budget they allocate to each country and maximize their welfare. In general, they can maximize their welfare in two ways: first, by optimally allocating their consumption choice between Home and Foreign goods within their allocated budgets and, second, by expanding their budget through their labor supply choice when labor supply is endogenous.

I next analyze the consumption side of boycotts in this setting with fixed labor supply.

2.3 The Consumption Side of Boycotts

Let $S(\mathbf{q}_i^B) \equiv \int_0^1 p_i(j) q_i^B(j) dj / (w_H l_H)$ be total expenditure as a share of domestic income of a representative boycotter on Home and Foreign goods, respectively.

Lemma 1 (Set of Boycotted Goods). *In any CE with boycotters where $\overline{L}_H < l_H$ or $e(\overline{L}_H) > 0$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0,1]; q_F^B(j) = 0\}$ is nonempty.*

This lemma implies that there is a complete prohibition on buying certain Foreign goods. More generally, the proof in the Appendix shows that boycotters choose two disjoint sets of Home and Foreign goods in which they will choose positive Home goods consumption or positive Foreign goods consumption.

In this setting, just as in the Dornbusch et al. (1977) economy, the patterns of trade for regular consumers are determined by a cutoff $\bar{j} = \bar{j}(\{w_H, w_F\})$ above which Foreign goods are imported, and let $S(\bar{j})$ denote the expenditure share of regular consumers on domestic goods. I further characterize the boycotted set of goods in the lemma below.

Proposition 2 (Optimal Set of Boycotted Goods)**.** *Let $\mathcal{J}_B$ be the optimal targeted set of boycotted goods given a level of tolerated Foreign welfare v^F:*

1. *There exists no set of positive measure between $\mathcal{J}_B$ and the trade cutoff $\bar{j}$, and $\mathcal{J}_B$ is a convex set.*
2. *There exists $j_B \in [\bar{j}; 1]$ so that $\mathcal{J}_B = [0, j_B]$ (almost everywhere) boycotts goods up to j_B.*

The proof of this proposition highlights what matters for the optimal set of boycotted goods. Domestic income from boycotted goods is proportional to the expenditure share on the set of boycotted goods, which I can loosely refer to as $S(\mathcal{J}_B)$. With CES preferences, this expenditure share $\bar{j}$ is the only determinant of the trade cutoff and Foreign wage because of the balanced trade equation, which equalizes domestic income with expenditure on Home-produced goods.

$$w_H l_H = \underbrace{(1-\lambda)S(\bar{j})w_H l_H}_{\text{regular domestic consumers}} + \underbrace{\lambda S(\mathcal{J}_B)w_H l_H}_{\text{domestic consumer boycotters}} + \underbrace{S(\bar{j})w_F l_F}_{\text{Foreign consumers}} \tag{11}$$

Any set of positive measure X between $\bar{j}$ and $\mathcal{J}_B$ or separating $\mathcal{J}_B$ can be transferred to the right of $\mathcal{J}_B$ to obtain a new set of boycotted goods with the same expenditure share and the same relative wage and trade cutoff as $\mathcal{J}_B$. Now, because the obtained set features boycotted goods of lower index j than $\mathcal{J}_B$, which are cheaper to produce for Home, welfare for Home boycotters would be higher under this alternative set of boycotted goods, while Foreign's welfare would remain unchanged. Thus, for any given utility v^F for Foreign, the optimal targeted boycotted set includes the goods closest to the trade cutoff.

2.4 The Labor Supply Side of Boycotts

I now turn to the labor supply side of boycotts. In our baseline DFS model with log utility of consumption, the income and substitution effects offset each other, so labor supply, even if endogenous, does not respond to wage changes.

I modify the DFS setup so that consumers choose consumption goods $\mathbf{q}$ and labor supply l_i to maximize

$$U(\mathbf{q}^i, l_i) = \int_0^1 s(j)\, u\big(q^i(j)\big)\, dj - v(l_i) \tag{12}$$

We consider CRRA utility for goods and an isoelastic disutility of labor:

$$u(q) = \frac{q^{1-\gamma} - 1}{1-\gamma} \quad \text{and} \quad v(l) = \frac{l^{1+\eta}}{1+\eta}, \tag{13}$$

with $1 \neq \gamma > 0$ and $\eta > 0$.

We can solve the boycotters' problem in two steps: First, they choose the consumption bundle across goods given income, and then they choose labor supply to maximize income. In the first step, Propositions 1 (equivalence of the CE with boycotters with fixed excess demand to the CE with boycotters with fixed expenditure) and 2 (optimal boycott set is a cutoff) generalize to these preferences, so we focus on the labor supply aspect of boycotts.

Proposition 3 (Labor Supply of Boycotters)**.** *In every optimal boycott, boycotters work more than regular consumers* $l^{B*} > l^*_H$.

To arrive at this proposition, observe that consumers' labor supply choice in the second step is determined by the indirect utility

$$V(w_i, l_i) = \frac{(w_i\, l_i)^{1-\gamma}\, P_i^{\gamma} - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta}. \tag{14}$$

where P_i is the price index faced by the consumer. The optimal labor supply of a boycotter is therefore

$$l^{B*} = \Big[P_B^{\gamma}\, w^{1-\gamma}\Big]^{\frac{1}{\eta+\gamma}}.$$

In effect, a boycotter faces a penalty in her consumption basket because she must pay more for the goods she buys domestically. Since $P_B > P$, $l^{B*} > l^*_H$.

To overcome the cost of having to buy domestically produced goods, the boycotter works more. In doing so, she earns more income; since her consumption is financed entirely by labor earnings, she allocates all her increased income to domestic goods. This additional spending on domestic varieties helps tilt the relative price in favor of Home production. In equilibrium, this upward pressure on domestic spending raises the domestic wage (and – through the cutoff condition that pins down production – lowers the trade cutoff $\overline{j}$ relative to the free–trade result), thereby harming Foreign's utility.

2.5 Symmetry and Leakage

Boycotts' Symmetric Effect on Imports and Exports. In the full-boycott-coalition economy, imports of foreign goods are

$$M_H = [1 - S(\mathcal{J}_b)]w_H l_H. \tag{15}$$

Thus, an expansion of the set of boycotted goods $\mathcal{J}_b$ directly decreases the share $1 - S(\mathcal{J}_b)$ and indirectly increases domestic wages due to increased demand for domestic goods. On net, the import share term dominates, and imports decrease.

Exports of domestic firm goods are

$$X_H = S(\bar{j}(j_B))w_F l_F. \tag{16}$$

An increase in the boycott cutoff j_B reduces the trade equilibrium cutoff $\bar{j}(j_B)$ and therefore decreases Home exports, which restores trade balance. Boycotts, therefore, distort both imports and exports. That is, a blanket boycott by all consumers is equivalent to a specific VER by domestic firms.

These adjustments are reminiscent of Lerner symmetry (Lerner (1936) and Costinot and Werning (2019)), which provides an equivalence between an import tariff and an export tax. Consider an economy A where everyone is a boycotter $\lambda = 1$ and the set of boycotted goods is $\mathcal{J}_B = [0, j_B]$. For the purposes of our analysis, this economy is the analog of an economy with an import tariff. Then consider an economy B where firms are barred from exporting any goods in $[j_R, 1]$, where j_R is an export restriction cutoff. This economy is the analog of an economy with an export tax. Then, economies A and B are equivalent if $j_R = \bar{j}(j_B)$.

Leakages from Nonboycotters. With the presence of nonboycotters $\lambda < 1$, a boycott $\mathcal{J}_B = [0, j_B]$ leads to the imports from nonboycotters equal to

$$M_{NB} = (1 - \lambda)[1 - S(\bar{j}(j_b))]w_H l_H. \tag{17}$$

The boycott indirectly raises domestic wages through reallocation toward domestic goods. In addition, we saw in (16) that the export share $1 - S(\bar{j}(j_b))$ increases when a boycott expands. The combination of the effect of boycotts on the domestic wage and the export share of nonboycotters raises the imports of nonboycotters M_{NB} from the foreign country. Therefore, a boycotter's unilateral attempts to prevent Foreign imports can be undermined by other households' responses to the wage and price changes she induces. This leakage is present in the literature on optimal unilateral behavior by consumers motivated by altruistic, non-individualistic preferences (see Kaufmann et al.

(2024) and Trammell (2025)). This equilibrium leakage pushes a boycotter to boycott a larger set of goods to tolerate the same level of Foreign welfare v^F, i.e., $j_B(\lambda, v^F)$, and Foreign importers per boycotted $M_{NB} = [1 - S(j_b)]w_H l_H$ are decreasing in λ.

2.6 Comparison with Trade Sanctions

Full Boycott Coalitions and Good-Specific Sanctions. To compare boycotts with sanctions, I now analyze the extreme cases of an economy where all Home consumers are boycotters and an economy with good-specific tariffs.

Consider our DFS economy when every domestic consumer is a boycotter ($\lambda = 1$). Start with the no-boycott equilibrium where j_B is the free-trade cutoff, which we denote $\bar{j}_{NB}$. Now, introduce a boycott $\mathcal{J}_B = [0, j_B]$, with cumulative expenditure share $S([0, j_B]) < 1$) and $j_B > \bar{j}_{NB}$.

In a DFS economy without boycotters ($\lambda = 0$), we can then construct good-specific import tariffs on goods indexed by $j \in [\bar{j}_{NB}, j_B)$, which replicate the equilibrium allocations (and trade flows) of the DFS economy with a boycott. The idea is to set good-specific tariffs for goods with indices between the free-trade cutoff $\bar{j}_{NB}$ and the boycott cutoff j_B high enough that no goods in $[\bar{j}_{NB}, j_B)$ are imported. Since no tariffs are collected in equilibrium, the shift in the trade balance (and export cutoff $\bar{j}(j_B)$) in the boycott economy and good-specific tariff economy are the same, and both arise from Home consumers allocating more toward Home goods. Because the BT condition for the full-boycott-coalition economy is

$$\frac{w_H \, l_H}{w_F \, l_F} = \frac{S(\bar{j}(j_B))}{1 - S(j_B)} \,. \tag{18}$$

the good-specific tariffs are constructed such that the new relative productivity curve crosses the BT curve at j_B:

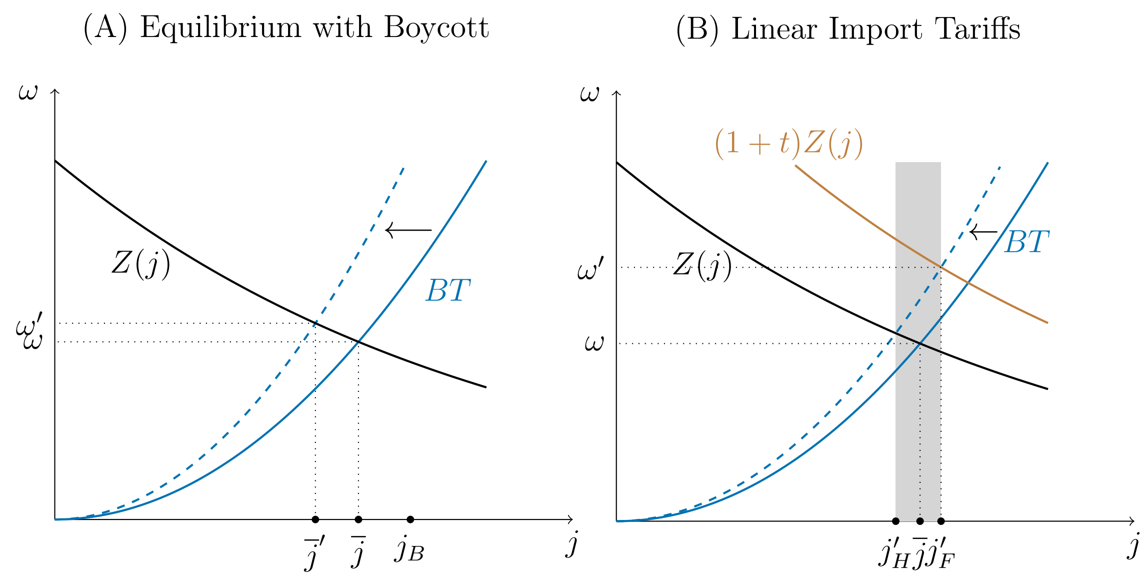
$$(1 + t(j))Z(j) > \frac{S(j)}{1 - S(j_B)} \text{ for all } j \in [\bar{j}_{NB}, j_B)] \text{ and } (1 + t(j_B))Z(j_B) = \frac{S(j_B)}{1 - S(j_B)} \tag{19}$$

Therefore, each boycott is equivalent to good-specific sanctions. In general, it is well known that any feasible allocation can be attained by good-specific and household-specific taxes. Thus, we need not limit ourselves to a comparison with good-specific tariffs alone to understand the link between boycotts and sanctions.

Comparison with Uniform Import Tariffs. I now compare the equilibrium with boycott to the equilibrium with a uniform import tariff. This comparison is instructive because Opp (2010), Costinot et al. (2015), and Becko (2022) show that optimal sanctions and terms-of-trade manipulations feature linear and uniform tariffs on imports.[3]

We can think of sanctions in the model with only regular domestic and Foreign customers (no boycotters) as a tariff t on Foreign exports q_F such that the domestic price of Foreign goods is $p_H(j) = p(j)(1 + t \cdot q_F)$ and revenues are rebated lump-sum to domestic consumers. Figure 1 illustrates the equilibrium with a boycott and the equilibrium with sanctions.

Figure 1: Equilibrium effects of an increase in the boycott cutoff and an increase in import sanctions



Notes: An increase in boycott cutoff j_B (left) and an increase in import t sanctions (right). An increase in the boycott cutoff shifts the BT curve to the left, and the equilibrium trade cutoff $\overline{j}$ (relative wage of Home workers ω) decreases (increases) to equal $\overline{j}'$ (ω'). An increase in import tariffs t shifts the relative productivity curve for imports to the right, and the transfers of collected tariffs lead to a slight leftward shift of the BT curve. Home exports goods $[0, j'_H]$, and Foreign exports $[j'_F, 1]$. The equilibrium relative wage of Home workers is higher; however, there is no trade in the shaded region.

First, in Panel B, sanctions make Foreign exports more expensive and shift the relative productivity curve (for Foreign exports) according to $\omega = (1 + t)Z(j)$, while boycotts keep the relative productivity curve unchanged. Since tariffs are rebated to domestic households, this shifts the balanced trade curve slightly to the left. These two forces on net lead to an increase in the relative wage of Home consumers and a decrease in the range of goods imported from Foreign for sanctions (in contrast to the increase in that range in the case of boycotts). In this DFS economy, the sanctions equilibrium features a set of nontraded goods $[j'_H, j'_F]$, while the full-boycott equilibrium features only a shift from the trade cutoff and relative wage from $(\overline{j}, \omega)$ to $\left(\overline{j}', \omega'\right)$.

[3]In general, I can treat sanctions as nonlinear tariffs $t(q_F)$ capturing a mixture of taxes and quotas and show that equilibrium can be implemented with a linear tax under complete information.

The correct intuition for consumer boycotts is, however, the economics of transfers when tastes differ geographically. The boycotters' income that would have been allocated to Foreign exports $\lambda w_H l_H S(\bar{j})$ can be interpreted as a unilateral transfer from Foreign to Home. Now, since these transfers would be consumed on domestic goods because of the difference in tastes between boycotters and Foreign consumers, this shifts the BT curve (and only the BT curve) to the left.

The loss for Foreign consumers stems from their inability to obtain goods previously imported from Home that Foreign workers are less productive at making. Even if, in a multicountry model, the Foreign country could switch to the second-best producer of those goods, it would still be worse off from the boycott. The gain for domestic regular consumers stems from the fact that trade is balanced. In a multicountry model, if trade were imbalanced and boycotters had to switch to the rest of the world rather than Home producers, then Home consumers would not be better off from the boycott.

3 Generalization and Discussion

3.1 A General Equivalence Result

I now prove our result in its most general setup. There are countries $i \in \mathcal{I}$, goods $g \in \mathcal{G}$, and households $h \in \mathcal{H}_i$ in each country i.

Production: There is one labor factor per country, elastically supplied by households. Each good g can be produced in each country i by a competitive firm with a constant-returns-to-scale (CRS) and strictly quasiconcave production function $F_{ig}(l_{ig}, M_{ig})$ that uses both the domestic labor factor and a composite of intermediate goods M_{ig}.

The non-substitution theorem (Samuelson, 1951) implies that prices are pinned down by the vector of wages in the world $\{w^i\}$, independent from demand, as are each firm's unit input and labor demands. This implies there exists (a) an exogenous function $p(\{w^i\})$ that, as a function of wages, gives the price of every good and (b) an exogenous consumption-to-labor-demand matrix-valued function $\mathcal{L}(\{w^i\})$ that, as a function of wages, gives a linear map from the vector of global consumption demands c to the vector of labor l demanded in each country.

Nonboycotters: Households have heterogeneous preferences $u^i(c, l; h)$. Nonboycotters choose their consumption basket and labor supply to maximize utility subject to their budget constraint.

$$\max_{c,l} u^i(c,l,h) \tag{20}$$

$$\text{s.t: } p(\{w^i\}) \cdot c \leq w^i l^i \tag{21}$$

Boycotters: Let $B_i \subseteq \mathcal{H}_i$ be the set of boycotting households in country i. Boycotters choose consumption and labor supply among those that hold fixed excess demand for each factor. I introduce some notation to formally define a boycotter's problem.

Let $c_g^{B_i} = \sum_{h \in B_i} c_g^i(h)$ the aggregate demand for good g by the boycotting coalition and c^{B_i} be the corresponding demand vector of the boycotting coalition. Recall from the non-substitution theorem that total labor factor demand arising from goods demand of boycotters $\mathcal{L}(\{w^i\})\, c^{B_i}$.

The boycotters' problem is to maximize their utility (20) subject to the budget constraint (21) and excess demand for factors being fixed at some $\overline{L} \in \mathbb{R}_+^{|\mathcal{I}|}$

$$l - \mathcal{L}(\{w^i\})\, c^{B_i} = \overline{L}. \tag{22}$$

Competitive Equilibrium: In a CE, firms price at marginal cost, nonboycotters and boycotters maximize their objective, and goods and labor markets clear.

$$(c^i(h), l^i(h)) \in \arg\max_{c,l} u^i(c,l;h) \quad \text{s.t.} \quad p(\{w^i\}) \cdot c^i(h) \leq w^i l^i(h) \quad \text{for all } i \in \mathcal{H}_i \setminus B_i$$

$$(c^i(h), l^i(h)) \in \arg\max_{c,l} u^i(c,l;h) \quad \text{s.t.} \quad \begin{cases} p(\{w^i\}) \cdot c^i(h) \leq w^i l^i(h) & \text{for all } i \in B_i \\ l - \mathcal{L}(\{w^i\})\, c^{B_i} = \overline{L}. \end{cases}$$

$$c_g = \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}_i} c_g^i(h) \quad \text{for all } g \in \mathcal{G} \tag{23}$$

$$l^i = \sum_{h \in \mathcal{H}_i} l^i(h) \quad \text{for all } i \in \mathcal{I} \tag{24}$$

$$l = \mathcal{L}(\{w^i\})c \tag{25}$$

Proposition 4. *(General Equivalence Result.) Any tuple* $\{c, M, l, w, \overline{L}\}$ *is a CE with boycotters with fixed excess demand for factors if and only if* $\{c, M, l, w, w \cdot (l - \overline{L})\}$ *is a CE with boycotters with fixed expenditures.*

This proposition shows that our main result holds in general in the setting where the non-substitution theorem applies with a unique factor in each country, labor, and

CRS production function. It arises from the fact that in the setting where the non-substitution theorem applies, with CRS technology and a unique labor factor in each country, there is an exogenous mapping from wages to factor prices that we construct from the consumption-to-labor-demand matrix. From this link, we can show that excess demand for factors is fixed at $\overline{L}$ if and only if boycotter expenditures for each country's goods are fixed at $w \cdot (l - \overline{L})$. In addition, from the system of constraints of our two equilibrium definitions, all choices from the CE with fixed excess and demand and the CE with fixed expenditures have the same effect on the world wage vector. Thus, these notions of CEs with boycotters are equivalent.

3.2 Discussion

Substitution Elasticities. Our baseline model is one in which the elasticity of substitution is equal to 1. Here, I consider different possible elasticities of substitution that can be captured in my framework by CES preferences:

$$U(\mathbf{q}) = \left(\int_0^1 s(j)^{\frac{1}{\gamma}} q^i(j)^{\frac{\gamma-1}{\gamma}} dj \right)^{\frac{\gamma}{\gamma-1}} \tag{26}$$

In this setting, whenever preferences are not Cobb–Douglas ($\gamma \neq 1$), the expenditure share on each good j depends not only on the exogenous $s(j)$ but also on the endogenous price of good j and the price index P faced by regular consumers and boycotters. In Appendix A.6, I show that our Proposition 2 on the optimal boycott set being determined by a cutoff generalizes to these preferences.

The case with heterogeneous elasticities of substitution across goods j is not analytically tractable. However, since consumers find it easier to switch to alternative products with a higher elasticity of substitution relative to that of other goods, I conjecture that if they decide to boycott, it is optimal to boycott goods close to $\overline{j}$ in relative productivity and substitutable with other goods.

Comparative Statics. The marginal change in the boycotters' welfare for a given decrease in Foreign's welfare captures the slope of the Pareto frontier. An increase in the boycott cutoff j_B yields a marginal decrease in Foreign's welfare. Home boycotters lose from consuming more expensive domestic goods but gain from higher relative wages. The loss from consuming more expensive domestic products is proportional to the elasticity of the expenditure share on boycotted goods that is shifted to domestic goods to the relative wage $\varepsilon_{S(j_B),\omega}$ (see Appendix A.7). Targeted boycotts are, therefore, more effective when this elasticity is smaller.

Good-Specific Boycotts. Boycotts often have a narrower target than a whole country. For instance, consumer groups can target a specific group of firms that may operate domestically or in Home because of the firms' political salience or animus toward the firms' actions. I can capture these motivations in my framework with the following preferences and choices for boycotters[4]:

$$U^B = \max_{\{\mathbf{q}^B\}} \int_0^1 s(j)(1 - d(j)) \log(q^B(j))dj \tag{27}$$

where the damage $d(j) \in [0; 1]$ captures the political salience of or animus toward firms producing good j, and $q^B(j)$ is the boycotters' consumption of good j irrespective of its country of production. In this setting, boycotters reduce their expenditure share on goods with high $d(j)$. In Appendix A.8, I show that if the damage to goods that Home imports is higher than the average damage, then the boycott improves the terms of trade for Home, while it harms the terms of trade if the damage to goods that Home exports is higher than the average damage.

Strategic Retaliation. Retaliation occurs when a coalition of Foreign consumers boycotts the exports of the home country in return. This situation leads to strategic considerations where the share of boycotters endogenously depends on the response of the foreign country. I model this in a setting where $j_B = 1$ and a share λ^F of Foreign boycotts all Home goods (the equivalent for Home boycotters is λ^H). The balanced trade condition then becomes

$$\omega = \frac{1 - \lambda^F}{1 - \lambda^H} \cdot \frac{S(\bar{j})}{1 - S(\bar{j})} \cdot \frac{l_F}{l_H} \tag{28}$$

We see that if each country chooses its optimum share of boycotters, then a prisoner's dilemma, reminiscent of the situation in a trade war, emerges: An increase in the boycotter share tilts the terms of trade in one's favor.[5]

For instance, for my application in the Eaton and Kortum (2002) specification with $z_F(j) = 1$ for all j, $z_H(j) = \left(\frac{T_H}{T_F}\right)^{\frac{1}{\mu}} \left(\frac{j}{1-j}\right)^{-\frac{1}{\mu}}$, with $T_H = T_F = 1$, $\mu = 1$, I show that the welfare of a representative Foreign nonboycotter is $\ln\left(\frac{1}{\omega} + 1\right)$. If the Foreign government cares only about the utility of its representative nonboycotter, the best response to a boycott λ^H is to increase λ^F arbitrarily close to 1. However, if both governments are utilitarian, the only Nash equilibrium is one where no country boycotts $\lambda^H = \lambda^F = 0$, as the welfare loss from boycotters going into autarky outweighs the terms-of-trade gains for nonboycotters. Therefore, it is difficult to justify boycotts on

[4]This setting corresponds to a situation where Home boycotters have different expenditure shares equal to $s'(j) = s(j)(1-d(j))$ that, importantly, do not necessarily sum to 1, so the preferences are not Cobb–Douglas.

[5]Opp (2010) studies tariff wars in a similar Ricardian model with a continuum of goods.

the basis of purely internal equity arguments.

4 Conclusion

This paper provides the first systematic analysis of international boycotts within a neoclassical Ricardian trade framework. My general equivalence result shows that equilibrium outcomes with boycotts can be equivalently analyzed through either fixed factor excess demands or fixed expenditures, providing conceptual clarity on the economics underlying boycott actions. Optimal boycotts prohibit certain Foreign goods, and boycotters only import goods for which Foreign's comparative advantage is sufficiently large. In addition, boycotters optimally work more to increase their budget to spend on domestically produced goods.

I show the equivalence between targeted consumer boycotts and certain forms of government-imposed sanctions or voluntary export restrictions (VERs). However, the effectiveness of boycotts is moderated by leakage effects from nonboycotting households and strategic retaliation risks from Foreign. Thus, while boycotts share important economic parallels with traditional trade policy tools, they also introduce unique complexities and can play a distinctive role as an instrument of geoeconomic influence.

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A Appendix

A.1 Proof of Equivalence Proposition 1

Starting from a competitive equilibrium (CE) with boycotters that hold fixed excess demand, we have excess demand for Home labor

$$l_H - \lambda \int_0^1 \frac{q_H^B(j)}{z_H(j)}\, dj = \overline{L}_H, \tag{29}$$

multiplying by the Home wage to make expenditures appear yields

$$w_H l_H - \lambda \int_0^1 \frac{w_H q_H^B(j)}{z_H(j)}\, dj = w_H \overline{L}_H. \tag{30}$$

Replacing the marginal product of labor at Home yields

$$w_H l_H - \lambda \int_0^1 p_H(j) q_H^B(j)\, dj = w_H \overline{L}_H. \tag{31}$$

Now, replacing the second term with the total expenditure on Home goods yields

$$w_H l_H - E(\mathbf{q}_H^B) = w_H \overline{L}_H. \tag{32}$$

All the steps taken can be reversed and are equivalences, that is, iff

$$E(\mathbf{q}_H^B) l_H = w_H \left(l_H - \overline{L}_H \right). \tag{33}$$

A similar equivalence obtains between fixed excess demand for Foreign labor and fixed Foreign expenditures. In addition, from the system of constraints of our two notions of CE, all such choices have the same effect on the world wage vector. Thus, the equivalence result.

A.2 Proof of Existence of Boycotted Goods Lemma 1

At prices

$$p_H(j) = \frac{w_H}{z_H(j)} \quad \& \quad p_F(j) = \frac{w_F}{z_F(j)}, \tag{34}$$

for interior $q_H^B(j)$ and $q_F^B(j)$ choices that do not saturate expenditure budgets, the first-order conditions for domestic boycotters are:

$$\frac{s(j)}{q_H^B(j)+q_F^B(j)} = \mu p_H(j) \quad \forall j \in [0,1] \quad \& \quad q_H^B(j) > 0 \tag{35}$$
$$\frac{s(j)}{q_H^B(j)+q_F^B(j)} = \mu p_F(j) \quad \forall j \in [0,1] \quad \& \quad q_F^B(j) > 0 \tag{36}$$

where μ is the Lagrangian on the boycotter's budget constraint and $\mu^{-1} = w_H l_H$ from summation and equalization with the budget constraint. Suppose by contradiction that (i) $\forall j \in [0,1] \quad q_F^B(j) > 0$. Suppose (ii) that there exists a set of positive measure Y such that $q_H^B(\cdot) > 0$ on Y. Since $p_H(\cdot)$ and $p_F(\cdot)$ cross at a unique point, then (35) and (36) cannot hold jointly on Y. That is, given (i), assumption (ii) is a contradiction. Therefore, on any set of positive measure, $q_H^B(\cdot) = 0$, and so the expenditure on domestic goods is zero. This contradicts the lemma's assumption. Thus, (i) is false. Therefore, if $e(\overline{L}_H) > 0$, the set of completely boycotted Foreign goods $\mathcal{J}_B = \{j \in [0,1]; q_F^B(j) = 0\}$ is nonempty. By symmetry, the only consistent choices with positive budgets are

$$p_H(j)q_H^B(j) = s(j)w_H l_H \quad \forall j \in [0,1] \quad \& \quad q_H^B(j) > 0 \tag{37}$$
$$p_F(j)q_F^B(j) = s(j)w_H l_H \quad \forall j \in [0,1] \quad \& \quad q_F^B(j) > 0 \tag{38}$$

Note that we have shown, in passing, that boycotters choose sets of Home and Foreign goods in which they will choose positive Home goods consumption or positive Foreign goods consumption.

A.3 Proof of Optimal Labor Supply Proposition 3

For a given level of labor l_i (and hence income $w_i l_i$), the problem of a consumer facing arbitrary prices $p(j)$ is standard and is

$$\max_{\{q^i(j)\}} \int_0^1 s(j)\frac{q^i(j)^{1-\gamma}-1}{1-\gamma}\,dj \quad \text{s.t.} \quad \int_0^1 p(j)q^i(j)\,dj = w_i\,l_i.$$

Introducing a Lagrange multiplier μ for the budget constraint, the Lagrangian is

$$\mathcal{L} = \int_0^1 s(j)\frac{q^i(j)^{1-\gamma}-1}{1-\gamma}\,dj - \mu\left[\int_0^1 p(j)q^i(j)\,dj - w_i\,l_i\right].$$

The first–order condition with respect to $q^i(j)$ is

$$\frac{\partial \mathcal{L}}{\partial q^i(j)}: \quad s(j)\,q^i(j)^{-\gamma} - \mu\,p(j) = 0.$$

Thus,

$$q^i(j)^{-\gamma} = \mu \, \frac{p(j)}{s(j)} \quad \Longrightarrow \quad q^i(j) = \left(\frac{s(j)}{\mu \, p(j)} \right)^{1/\gamma}.$$

To pin down μ, substitute the solution into the budget constraint:

$$\int_0^1 p(j) \left(\frac{s(j)}{\mu \, p(j)} \right)^{1/\gamma} dj = w_i \, l_i.$$

This can be rewritten as

$$\left(\frac{1}{\mu} \right)^{1/\gamma} \int_0^1 s(j)^{1/\gamma} \, p(j)^{1-1/\gamma} \, dj = w_i \, l_i.$$

It is convenient to define a price index

$$P \equiv \int_0^1 s(j)^{\frac{1}{\gamma}} \, p(j)^{\frac{\gamma-1}{\gamma}} \, dj. \tag{39}$$

Then, the above condition reads

$$\left(\frac{1}{\mu} \right)^{1/\gamma} P = w_i \, l_i,$$

so that

$$\mu = \left(\frac{P}{w_i \, l_i} \right)^{\gamma}.$$

Plug this back into the optimal demand:

$$q^i(j) = \left(\frac{s(j)}{p(j)} \right)^{1/\gamma} \frac{w_i \, l_i}{P},$$

Plug the optimal consumption back into the utility from consumption. The consumption part of utility becomes

$$\int_0^1 s(j) \frac{q^i(j)^{1-\gamma} - 1}{1-\gamma} \, dj.$$

Further simplification of this expression yields

$$\frac{(w_i \, l_i)^{1-\gamma}}{1-\gamma} \, P^{\gamma-1} \, P = \frac{(w_i \, l_i)^{1-\gamma} \, P^{\gamma}}{1-\gamma}.$$

Subtracting the disutility of labor, the household's indirect utility is

$$V(w_i, l_i) = \frac{(w_i \, l_i)^{1-\gamma} \, P^{\gamma} - 1}{1-\gamma} - \frac{l_i^{1+\eta}}{1+\eta}.$$

Now, the household chooses l_i to maximize $V(w_i, l_i)$. Differentiate with respect to l_i:

$$\frac{\partial V}{\partial l_i} = P^{\gamma}(w_i)^{1-\gamma}\, l_i^{-\gamma} - l_i^{\eta} = 0.$$

Thus, the first–order condition is

$$P^{\gamma}(w_i)^{1-\gamma} = l_i^{\eta+\gamma}.$$

It follows that the optimal labor supply is given by

$$l_i^* = \left[P^{\gamma}(w_i)^{1-\gamma}\right]^{\frac{1}{\eta+\gamma}}.$$

Now, consider a domestic boycotter. Because she rejects Foreign varieties on $\mathcal{J}_B$ and buys only domestic varieties at price $p_H(j)$ on that set, her effective price index is

$$P_B = \int_0^1 s(j)^{\frac{1}{\gamma}}\, p_B(j)^{\frac{\gamma-1}{\gamma}}\, dj \tag{40}$$

where $p_B(j) = p_H(j)$ for boycotted goods $j \leq j_B$ and $p_B(j) = p(j)$, the world price, for non-boycotted goods.

However, since domestic prices $p_H(j)$ exceed the free–trade prices $p(j)$ in $[\bar{j}, j_B]$, we have $P_B > P$.

Therefore, $l^{B*} > l_H^*$.

A.4 Proof of General Equivalence Proposition 4

Consider a competitive firm in country i producing good g with a constant-returns-to-scale (CRS) production function $F_{ig}(l_{ig}, M_{ig})$. A convenient way to formulate the firm's choices is to consider the problem:

$$\min_{l_{ig}, M_{ig}} \; w^i\, l_{ig} + p_{m,ig}\, M_{ig} \quad \text{subject to} \quad F_{ig}(l_{ig}, M_{ig}) = 1,$$

which defines the unit cost function

$$c_{ig}(w^i, p_{m,ig}) = \min\{w^i\, l_{ig} + p_{m,ig}\, M_{ig} : F_{ig}(l_{ig}, M_{ig}) = 1\}.$$

Let $w \in \mathbb{R}^{|\mathcal{I}|}$ be the vector of wages across countries and $\mathcal{L}(\{w^i\}) \in \mathbb{R}_+^{|\mathcal{I}|\times|\mathcal{G}|}$ be the consumption-to-labor-demand matrix. By Shephard's lemma, the partial derivative of the unit cost function with respect to the wage gives the factor (labor) cost share:

$$\frac{\partial c_{ig}(w^i, p_{m,ig})}{\partial w^i} = [\mathcal{L}(\{w^i\})]_{ig},$$

where $[\mathcal{L}(\{w^i\})]_{ig}$ represents the amount of domestic factor used per unit of good g in country i.

In equilibrium, assuming free trade (or cost equalization across countries), the price of good g is given by its unit cost. Hence, the vector of goods prices $p \in \mathbb{R}^G$ can be written as a weighted sum of factor prices:

$$p_g = \sum_{i \in \mathcal{I}} w^i \, [\mathcal{L}(\{w^i\})]_{ig},$$

or in vector notation,

$$p = \mathcal{L}(\{w^i\})^\top w, \tag{41}$$

Define the excess demand (or the factor content of boycotters' consumption) as

$$\overline{L} \equiv l - \mathcal{L}(\{w^i\}) \, c^{B_i}$$

Multiplying both sides of this equation by the wage vector w (taking the dot product) gives:

$$w \cdot \overline{L} = w \cdot \Big[l - \mathcal{L}(\{w^i\}) \, c^{B_i} \Big].$$

From the production side (41), we can make expenditures appear by developing the expression and substituting in prices.

$$w \cdot \Big[l - \mathcal{L}(\{w^i\}) \, c^{B_i} \Big] = w \cdot l - \Big(\mathcal{L}(\{w^i\})^\top w \Big) \cdot c^{B_i} = w \cdot l - p \cdot c^{B_i}.$$

Therefore, we obtain:

$$p \cdot c^{B_i} = w \cdot \big(l - \overline{L} \big) \, .$$

In addition, from the system of constraints of our two notions of CE, all such choices have the same effect on the world wage vector. Thus, the equivalence result.

A.5 Proof of Proposition 2

Consider the set of Foreign boycotted goods $\mathcal{J}_B$. Then, domestic income is:

$$w_H l_H = \underbrace{(1-\lambda) \int_0^{\bar{j}} p(j) q^H(j) dj}_{\text{regular domestic consumers}} + \underbrace{\lambda \int_{\mathcal{J}_\mathcal{B}} p_H(j) q_H^B(j) dj}_{\text{domestic consumer boycotters}} + \underbrace{\int_0^{\bar{j}} p(j) q^F(j) dj}_{\text{Foreign consumers}} \tag{42}$$

Summing the expenditures as previously, we obtain

$$w_H l_H = (1-\lambda) S(\bar{j}) w_H l_H + \lambda \left(\int_{\mathcal{J}_\mathcal{B}} s(j) dj \right) w_H l_H + S(\bar{j}) w_F l_F \tag{43}$$

Denote the expenditure share in the set of boycotted goods $S(\mathcal{J}_B) \equiv \int_{\mathcal{J}_B} s(j)dj$ for ease of notation. Simplifying the equation above yields

$$\omega = \frac{S(\bar{j})}{1 - \lambda S(\mathcal{J}_B) - (1-\lambda)S(\bar{j})} \cdot \frac{l_F}{l_H} \tag{44}$$

Using the demand functions and equilibrium prices, we can integrate to find the total utility for the representative consumer in the foreign country:

$$U^F = \int_0^1 s(j) \log\left(\frac{s(j) w_F l_F}{p(j)}\right) dj \tag{45}$$

where the price function is given by

$$p(j) = \begin{cases} \frac{w_H}{z_H(j)} & \text{if } j \leq \bar{j} \\ \frac{w_F}{z_F(j)} & \text{if } j > \bar{j} \end{cases} \tag{46}$$

i.e., the Foreign real wage is

$$\frac{w_F}{p(j)} = \begin{cases} \frac{z_H(j)}{\omega} & \text{if } j \leq \bar{j} \\ z_F(j) & \text{if } j > \bar{j} \end{cases} \tag{47}$$

and welfare is

$$U^F = \int_0^{\bar{j}} s(j) \log\left(\frac{z_H(j)}{\omega}\right) dj + \int_{\bar{j}}^1 s(j) \log\left(z_F(j)\right) dj + \int_0^1 s(j) \log\left(s(j) l_F\right) dj \tag{48}$$

Observe here that Foreign's welfare depends only on $\mathcal{J}_B$ insofar as it affects the relative wage ω and the trade cutoff $\bar{j}$. However, the welfare of Home boycotters depends directly on $\mathcal{J}_B$ in that boycotted goods with higher index value j reduce the welfare of Home boycotters (they are more expensive to produce). Now, suppose that there exists a set of positive measure X between $\bar{j}$ and $\mathcal{J}_B$ or separating $\mathcal{J}_B$ ($\bar{j} < \inf X < \sup X < \inf \mathcal{J}_B$ or $\bar{j} < \sup X < \sup \mathcal{J}_B \quad \& \quad X \cup \mathcal{J}_B = \emptyset$). Then, by replacing a set with the same expenditure share as X from the right in $\mathcal{J}_B$ with X, we obtain a new set of boycotted goods with the same expenditure share and, from (43), the same relative wage and trade cutoff as $\mathcal{J}_B$. Therefore, from (48), this set gives the same utility to Foreign consumers v^F. Now, note that because this set features boycotted goods of a lower index value j than $\mathcal{J}_B$, Home boycotters would have higher welfare. Thus, a set $\mathcal{J}_B$ that does not satisfy Proposition 2 cannot be optimal.

A.6 Substitution Elasticities

For $\gamma \neq 1$, we can define the trade price index and boycotter price index

$$P(\gamma) = \left(\int_0^1 s(j) p(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}} \tag{49}$$

$$P_B(\gamma) = \left(\int_0^1 s(j) p_H(j)^{1-\gamma} dj \right)^{\frac{1}{1-\gamma}} \tag{50}$$

where $p(j)$ is the world price of good j and $p_H(j)$ is the domestic producer price of good j. Then, the expenditure shares on good j for regular consumers and boycotters are, respectively,

$$\text{share}(j;\gamma) = s(j) \left(\frac{p(j)}{P(\gamma)} \right)^{1-\gamma} \tag{51}$$

$$\text{share}^B(j;\gamma) = s(j) \left(\frac{p_H(j)}{P_B(\gamma)} \right)^{1-\gamma} \tag{52}$$

Aggregating these shares,

$$S(j;\gamma) = \int_0^j \text{share}(j';\gamma) dj' \tag{53}$$

$$S^B(j;\gamma) = \int_0^{j'} \text{share}^B(j';\gamma) dj' \tag{54}$$

Domestic income is

$$w_H l_H = (1-\lambda) S(\bar{j};\gamma) w_H l_H + \lambda S^B(j_B;\gamma) w_H l_H + S(\bar{j};\gamma) w_F l_F \tag{55}$$

We obtain the new BT condition:

$$\omega = \frac{S(\bar{j};\gamma)}{1 - \lambda S^B(j_B;\gamma) - (1-\lambda) S(\bar{j};\gamma)} \cdot \frac{l_F}{l_H} \tag{56}$$

We see that we obtain the same balanced trade condition as before with expenditure shares that are endogenous. Thus, the same analysis as for Propositions 1 and 2 and Lemma 1 holds.

A.7 Comparative Statics

Proposition 5. *A reduction in $v^F \downarrow$ comes with an increase in the optimal boycott cutoff ($j_B \uparrow$), increases the relative wage of Home workers $\omega \uparrow$, and increases the set of*

Foreign-produced goods $\bar{j} \downarrow$. In particular, the marginal welfare changes are:

$$\frac{dv^F}{dj_B} = -S(\bar{j})\frac{d\omega}{\omega dj_B} \leq 0 \tag{57}$$

and the marginal change in Home boycotters' welfare relative to the change in Foreign's welfare is

$$\frac{dU^B}{-dv^F} = \frac{1}{S(\bar{j})} \cdot \left(1 - S(j_B) + S(j_B) \cdot \log\left(\underbrace{\frac{Z(j_B)}{\omega}}_{<1}\right)\varepsilon_{S(j_B),\omega}\right) \tag{58}$$

where the elasticity is:

$$\varepsilon_{S(j_B),\omega} \equiv \frac{d\log S(j_B)}{d\log\omega} = \frac{1 - S(\bar{j}) - \lambda(S(j_B) - S(\bar{j}))}{\lambda S(j_B)}\left(1 - \frac{\frac{(1-\lambda S(j_B))s(\bar{j})\omega}{1-S(\bar{j})-\lambda(S(j_B)-S(\bar{j}))}}{\frac{(1-\lambda S(j_B))s(\bar{j})\omega}{1-S(\bar{j})-\lambda(S(j_B)-S(\bar{j}))} - Z'(\bar{j})}\right)^{-1} \tag{59}$$

Proof. Consider j_B, the cutoff above which Foreign goods are boycotted. Domestic income is

$$w_H l_H = \underbrace{(1-\lambda)\int_0^{\bar{j}} p(j)q^H(j)dj}_{\text{regular domestic consumers}} + \underbrace{\lambda\int_0^{j_B} p_H(j)q_H^B(j)dj}_{\text{domestic consumer boycotters}} + \underbrace{\int_0^{\bar{j}} p(j)q^F(j)dj}_{\text{Foreign consumers}} \tag{60}$$

Summing the expenditures as previously, we obtain

$$w_H l_H = (1-\lambda)S(\bar{j})w_H l_H + \lambda S(j_B)w_H l_H + S(\bar{j})w_F l_F \tag{61}$$

Simplifying the equation above yields:

$$\omega = \frac{S(\bar{j})}{1 - \lambda S(j_B) - (1-\lambda)S(\bar{j})} \cdot \frac{l_F}{l_H} \tag{62}$$

From here, we see that an increase in the boycott cutoff j_B reallocates expenditure toward domestic goods and shifts the balanced trade curve (62) to the left. The equilibrium relative wage of domestic workers increases, and the trade equilibrium cutoff decreases.

The welfare of each Home boycotter is

$$U^B = \int_0^{j_B} s(j)\log\left(z_H(j)\right)dj + \int_{j_B}^1 s(j)\log\left(z_F(j)\omega\right)dj + \int_0^1 s(j)\log\left(s(j)l_H\right)dj \tag{63}$$

We obtain

$$\frac{dU^B}{dv^F} = (1 - S(j_B))\,\frac{d\omega}{\omega dv^F} + s(j_B)\log\left(\underbrace{\frac{z_H(j_B)}{\omega \cdot z_F(j_B)}}_{<1}\right)\frac{dj_B}{dv^F} \tag{64}$$

Since

$$\frac{dv^F}{d\omega} = -S(\bar{j})\frac{1}{\omega} \tag{65}$$

we obtain

$$\frac{dU^B}{dv^F} = -\frac{(1 - S(j_B))}{S(\bar{j})} - s(j_B)\log\left(\underbrace{\frac{z_H(j_B)}{\omega \cdot z_F(j_B)}}_{<1}\right)\frac{\omega \cdot dj_B}{S(\bar{j})d\omega} \tag{66}$$

From the implicit function theorem, we have:

$$\frac{d\omega}{dj_B} = \frac{\lambda s(j_B)\omega}{1 - \lambda S(j_B) - (1-\lambda)S(\bar{j})}\left(1 - \frac{\frac{(1-\lambda S(j_B))s\omega}{S(1-\lambda S(j_B)-(1-\lambda)S)}}{\frac{(1-\lambda S(j_B))s\omega}{S(1-\lambda S(j_B)-(1-\lambda)S)} - Z'(\bar{j})}\right) \tag{67}$$

where $S = S(\bar{j})$. □

A.8 Good-Specific Boycotts

This setting corresponds to a situation where Home boycotters have different expenditure shares equal to $s'(j) = s(j)(1 - d(j))$ that, importantly, do not necessarily sum to 1, so the preferences are not Cobb–Douglas. Home boycotters' consumption choices are determined by the first-order conditions:

$$p(j)q_H^B(j) = s(j)\frac{1 - d(j)}{1 - D}w_H l_H \quad \forall j \in [0, 1] \tag{68}$$

where the average damage is $D \equiv \int_0^1 s(j)d(j)dj$ and we define $D(j) \equiv \int_0^j s(j')d(j')dj'$. Thus, Home boycotters reduce their expenditure share in goods according to how $d(j)$ compares to the average damage. Domestic income is

$$w_H l_H = (1-\lambda)S(\bar{j})w_H l_H + \lambda\frac{S(\bar{j}) - D(\bar{j})}{1 - D}w_H l_H + S(\bar{j})w_F l_F \tag{69}$$

The BT condition is therefore

$$\omega = \frac{S(\bar{j})}{1 - S(\bar{j}) - \lambda \left(\frac{S(\bar{j})D - D(\bar{j})}{1-D} \right)} \cdot \frac{l_F}{l_H} \tag{70}$$

Denote by $\bar{j}_{NB}$ the trade cutoff without any boycotters, for which the BT condition is the standard Dornbusch et al. (1977) one:

$$\omega = \frac{S(\bar{j}_{NB})}{1 - S(\bar{j}_{NB}))} \cdot \frac{l_F}{l_H} \tag{71}$$

Then, from equation (70), if $D(\bar{j}_{NB}) > D$, there is a rightward shift in the balanced trade curve that crosses $Z(j)$ after $\bar{j}_{NB}$, and the trade cutoff equilibrium with boycotters is such that $\bar{j} > \bar{j}_{NB}$ and $\omega < \omega_{NB}$. The situation is reversed if $D(\bar{j}_{NB}) < D$. There is a rightward shift in the balanced trade curve that crosses $Z(j)$ before $\bar{j}_{NB}$, and the trade cutoff equilibrium with boycotters is such that $\bar{j} < \bar{j}_{NB}$ and $\omega > \omega_{NB}$.